A Prior-Informed Bootstrap Framework for Network Credible Intervals Under Non-Random Missingness

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Generalizing methods from Daniele Bellutta, *Forensic Analysis of Observed Social Networks*

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Abstract

This document specifies a bootstrap-based framework for constructing approximate Bayesian credible intervals on network metrics computed from incomplete network observations. The framework extends and addresses limitations of the bootstrap methods described in Chapters 3 and 4 of Bellutta's dissertation (*Forensic Analysis of Observed Social Networks*) by introducing three key innovations: (1) explicit modeling of node-level missingness with a user-supplied prior on the correlation between centrality and missingness; (2) structural-role binning via the E/I index to recognize brokers as distinct from clustered hubs of equal degree; and (3) rank-conditional probabilistic imputation of edges involving nominated non-respondents, generalizing the global reciprocity-rate approach used in Smith et al. (2022) (the present author's prior work) to a rank-rank conditional reciprocity matrix. These extensions allow the framework to handle unweighted networks (which the prior framework could not), networks with distance-semantic weights (via a transformation step), and networks whose structure departs from the stylized models used in prior evaluations. This document provides the algorithmic specification in full detail, intended as the basis for Python implementation and subsequent empirical validation.

1 Introduction

Network analysts routinely work with incomplete observations of an underlying true network. Whether the data come from surveys with non-response, surveillance with limited resources, social media APIs with rate limits, or informant testimony about covert organizations, the observed network G_{obs} is typically only a partial view of some unobserved true network T. Quantifying uncertainty about network measures computed on G_{obs} has long been a difficult problem, particularly when the missingness mechanism is non-random and varies systematically with node properties such as centrality.

This framework provides a principled approach to constructing credible intervals on network metrics in this setting. The user supplies:

- An observed network G_{obs}
- A prior $\rho \in [-1, 1]$ on the correlation between node centrality and probability of being missing from observation
- Their best estimate of the magnitude of missingness (π_{total}, or the components π_{node} and π_{edge})
- The network measure(s) of interest

The framework then generates a posterior sample of plausible true networks consistent with these inputs, computes the requested measures on each, and returns credible intervals derived from the posterior sample.

The motivating use cases include covert network analysis (where central actors invest heavily in concealment), survey-based network studies (where non-response often correlates with status or visibility), social media analysis (where streaming APIs systematically over-represent active users), and any setting where the analyst has domain knowledge about the observation process but cannot fully characterize it. In each case, the framework's prior ρ allows the analyst to encode this knowledge directly, and the resulting credible intervals reflect both the magnitude and the structure of the assumed missingness.

2 Relationship to Bellutta's Prior Work

The framework builds on the bootstrap methods described in Chapters 3 and 4 of Bellutta's dissertation, but extends them in several important ways. Understanding these extensions requires briefly recapping the prior framework's scope and limitations.

2.1 Chapter 3: The Underlying Noise Model

Bellutta's Chapter 3 introduced a noise model used to simulate missing data on known true networks for the purpose of evaluating link-inference algorithms. The mechanism removes integer edge-weight units uniformly at random across all weight units in the network — equivalent to sampling edges with probability proportional to current weight and decrementing by one, with replacement. The total amount removed is a specified percentage of total edge weight.

This mechanism has several important properties:

- It is MCAR at the level of weight units, not at the level of edges
- High-weight edges lose more weight in expectation but rarely vanish
- No new edges are created during the removal process
- The node set is preserved entirely — only edges and weights change
- It assumes the network is weighted; the mechanism has no analog for unweighted networks

2.2 Chapter 4: The Bootstrap Framework

Bellutta's Chapter 4 used Chapter 3's noise model as the basis for a bootstrap approach to constructing confidence intervals. Given an observed incomplete network, five methods were evaluated for generating replicate networks: random augmentation (adding integer weight units uniformly at random), the random dot product graph (RDPG) bootstrap, the masked RDPG bootstrap, the structural perturbation method (SPM), and a non-negative matrix factorization perturbation method (NMFP). Confidence intervals were constructed by computing the metric of interest on each replicate and forming a normal-approximation interval centered at the observed metric value.

The Chapter 4 framework has several limitations that motivate this work:

1. **It assumes MCAR.** All five bootstrap methods operate under the assumption that missingness is uniform across edge-weight units. None can encode prior beliefs about which edges or nodes are more likely to be missing.

2. **It cannot handle unweighted networks.** Because Chapter 3's noise model removes integer weight units, the framework cannot operate on networks where edges are simply present or absent. An entire class of common networks falls outside its scope.

3. **It cannot handle distance-semantic weights.** The noise model assumes weights represent counts where higher equals stronger. Distance networks, where higher weight means weaker tie, would be processed incorrectly.

4. **It does not model node-level missingness.** Chapter 3 removes weight from edges but preserves the node set. In practice, much network missingness is at the node level: non-respondents in surveys, unobserved actors in covert networks, accounts excluded from a social media sample. The Chapter 4 framework has no mechanism for this.

5. **It was evaluated only on stylized networks.** The framework's empirical validation relied on uniformly random, small-world, and scale-free network models. Real networks often combine structural features in ways these models do not capture, and the framework's behavior on such networks is unclear.

6. **Empirical results showed intervals were often miscalibrated.** Chapter 4's coverage results showed that the bootstrap methods produced intervals that were either too narrow (insufficient coverage) or too wide to be useful. Improvements in both the underlying missingness model and the structural assumptions are warranted.

2.3 What This Framework Adds

The framework introduced in this document addresses each limitation:

Explicit node-level missingness. The framework treats missing nodes as a first-class mechanism, separate from edge-weight loss. The parameter π_{node} governs the fraction of nodes believed missing, and a synthetic-node augmentation step adds plausible missing nodes to each replicate. The retained edge-weight mechanism (π_{edge}) becomes optional residual error on observed nodes.

Centrality-correlated missingness via a user-supplied prior. The parameter $\rho \in (-1, 1)$ encodes the user's domain-knowledge belief about the association between node centrality and probability of being missing from observation. Positive ρ corresponds to settings where central actors are more likely to be hidden (covert networks, non-response among prominent people); negative ρ corresponds to settings where central actors are more visible (social media, surveillance prioritizing known figures); zero ρ recovers MCAR behavior. The framework converts ρ into a bin-assignment distribution that biases the centrality of added nodes accordingly.

Internally, ρ is interpreted as Kendall's τ_b, the rank-based correlation between the missingness indicator (binary) and node centrality (continuous). This choice matches the user's rank-order intuition about missingness ("low-ranked nodes are likely to be missing" is a statement about ranks, not raw centrality values), and it avoids an asymmetric-ceiling problem that arose under Pearson correlation in early framework development. On heavy-tailed networks (e.g., Marvel co-appearance, citation networks), Pearson correlation between missingness and centrality saturates at a much lower magnitude on the negative side than the positive side, because the many low-centrality nodes have near-identical values that fail to discriminate missing from observed. Kendall τ_b on ranks produces symmetric ceilings across networks and across the sign of ρ, making the framework's behavior consistent regardless of network structure. The user-facing parameter $\rho \in (-1, 1)$ is unchanged from a standard correlation interface; the metric switch is internal.

Native support for unweighted networks. Because node-level missingness is independent of edge-weight semantics, the framework operates on unweighted networks by skipping the edge-weight reversal step entirely. This is a substantive extension: the Chapter 4 framework cannot handle unweighted networks at all, but unweighted networks are central to several lines of prior work on missing-data bias in survey-based networks (e.g., Smith et al., 2022, of which the present author was a co-author), and supporting them is one of the primary motivations for treating node missingness explicitly.

Distance-semantic weight transformation. The framework provides automatic transformation of distance-weighted networks to a count-like scale before processing, with three transformation methods (scaled reciprocal, max-minus-distance, exponential decay). Credible intervals returned by the framework apply to the transformed network, with explicit caveats about back-transformation.

Reduced reliance on stylized network models. Rather than fitting a parametric generative model (e.g., RDPG) to the observed network and bootstrapping from it, the framework uses empirical rank-rank connectivity directly. Each replicate's added-node edges are sampled from the observed network's empirical structure, conditional on the centrality and brokerage bins involved. The framework therefore makes no assumption that the true network follows any particular stylized form — it only assumes that the rank-rank structure observed in G_{obs} is representative of the true network's rank-rank structure.

Structural-role binning via the E/I index. The framework uses two binning dimensions by default: degree centrality rank and the E/I index. The E/I index, derived from Louvain community detection on G_{obs}, distinguishes brokers (whose ties cross community boundaries) from clustered hubs (whose ties remain within a single community). This addresses a key weakness of degree-only binning: two nodes with identical degree may play fundamentally different structural roles, and treating them as exchangeable miscalibrates credible intervals for measures sensitive to bridging behavior (e.g., betweenness centrality). The framework falls back to degree-only binning automatically when community structure is not meaningful.

Rank-conditional probabilistic imputation for nominated non-respondents. In survey, surveillance, and informant-based settings, the analyst frequently knows that certain unobserved nodes exist because they appear as targets of edges from observed (respondent) nodes. These "nominated non-respondents" carry partial information — their incoming edges are known — but their outgoing behavior must be imputed. Smith et al. (2022), the present author's prior work on missing-data bias in survey-based networks, found that probabilistic imputation, using the global reciprocity rate to impute reverse-direction edges, worked well across many directed-network settings. The framework generalizes this approach by computing a rank-conditional reciprocity matrix R where each cell gives the reciprocity rate for forward edges between specific rank-rank cell pairs, rather than a single global rate. This captures structural variation in reciprocity behavior (e.g., peripheral nodes reciprocate at different rates than central hubs, brokers exhibit characteristic asymmetries) while remaining computationally tractable, in contrast to model-based imputation approaches (e.g., ERGMs) that scale poorly. The framework distinguishes nominated non-respondents (handled via R-based imputation in a new Stage 0.5) from non-nominated non-respondents (handled via the rank-rank matrix P in Stage 1).

Bayesian framing. The framework's output is an approximate posterior sample over plausible true networks, conditional on the observed network and the user's prior. Credible intervals are derived directly from quantiles of the posterior sample, without the normal-approximation assumption of Chapter 4. The framework is not derived from an explicit likelihood and prior density formulation, so its output is "approximately" Bayesian in spirit rather than strictly so, but the conceptual framing is fundamentally different from Chapter 4's frequentist confidence intervals.

3 Algorithm Specification

This section provides the complete algorithmic specification of the framework. The algorithm proceeds in three phases: a setup phase executed once on the observed network, a replicate generation phase executed B times to produce the posterior sample, and an aggregation phase that summarizes the posterior.

3.1 Inputs

Required inputs.

- G_{obs}: an observed network with N nodes. The network may be directed or undirected, weighted (with non-negative integer edge weights representing counts) or unweighted. Distance-semantic networks must be transformed (see Section 3.3).
- $\rho \in [-1, 1]$: the prior correlation between node centrality and probability of being missing.
- One or more network measures m for which credible intervals are desired.

Missingness specification. The user provides either the simple or advanced interface:

- Simple: $\pi_{\text{total}} \in [0, 1]$, the overall fraction of the true network believed unobserved. The framework defaults this to $\pi_{\text{node}} = \pi_{\text{total}}$, $\pi_{\text{edge}} = 0$.
- Advanced: $\pi_{\text{node}} \in [0, 1]$ and/or $\pi_{\text{edge}} \in [0, 1]$, the node-level and edge-level missingness fractions respectively. If only one is supplied, the other defaults to 0.

At least one of π_{node} or π_{edge} must be positive. For unweighted networks, π_{edge} is structurally inapplicable and must be 0; $\pi_{\text{node}} > 0$ is required.

Sign convention for ρ.

- $\rho > 0$: central nodes are *more* likely to be missing (e.g., they hide successfully, refuse to respond, are hard to reach)
- $\rho < 0$: central nodes are *less* likely to be missing (e.g., they are prioritized for surveillance, naturally over-captured by streaming APIs)
- $\rho = 0$: no correlation between centrality and missingness (MCAR)

A negative ρ does not denote a negative probability; it denotes a negative correlation between centrality and the indicator of being missing.

The framework interprets ρ internally as Kendall's τ_b, the rank-based correlation between centrality and the missingness indicator. The user-facing parameter is a standard correlation in $(-1, 1)$; the rank-based interpretation is internal and not exposed to the user except in diagnostic output. See Section 3.2 for the feasibility constraints that determine whether a given ρ is structurally achievable on a given network at a given missingness rate.

Guidance table for ρ. Table 1 provides illustrative values of ρ for common observation processes. Users should set ρ based on domain knowledge, not by estimating it from G_{obs}.

Observation process	**Direction**	**Typical $\|\rho\|$**	**Signed ρ**
Complete-roster survey, high response rate	No bias	0.0	0.0
Social media via streaming/firehose API	Central more visible	0.3 – 0.5	-0.3 to -0.5
Snowball sample from initial respondents	Central more visible	0.4 – 0.6	-0.4 to -0.6
Survey with refusals among prominent people	Central more hidden	0.3 – 0.5	$+0.3$ to $+0.5$
Covert network from informant testimony	Central more hidden	0.4 – 0.7	$+0.4$ to $+0.7$
Covert network from police surveillance	Central more visible	0.5 – 0.7	-0.5 to -0.7
Mixed police data	Variable	Variable	User judgment

Table 1: Guidance for setting ρ based on observation process. Values are illustrative starting points, not prescriptions. Whether a chosen ρ is structurally achievable on a given network at a given missingness rate is a separate question; the framework surfaces this via `feasible_rho_range` (Section 3.2). On heavy-tailed networks at low missingness rates, the rate-bounded ceiling $\sqrt{2p(1-p)}$ may limit the achievable $|\rho|$ below the values in this table; the framework will then operate at the practical ceiling and report both nominal and realized values.

Optional inputs.

- `community_labels`: user-supplied community assignments, overriding Louvain detection
- `weight_semantics`: `"count"` (default) or `"distance"`; selects transformation behavior
- `partially_observed_nodes`: a list of node identifiers for nodes that appear in G_{obs} as edge targets (i.e., they have incoming edges from respondents) but did not respond themselves. These are the "nominated non-respondents" handled by Stage 0.5. If not supplied, all node-level missingness is assumed to involve non-nominated nodes (handled by Stage 1).

Algorithm hyperparameters with defaults.

- B: number of posterior samples (default 1000)
- K: number of degree-rank bins. Default is automatic selection (`K = :auto`) via the function `find_optimal_K`, which iterates K from a maximum tractable value down to a floor and returns the largest K that satisfies the framework's three algorithmic priors (see Section 3.11). Larger K produces a Kendall τ_b ceiling closer to the rate-bounded absolute (fewer bin-level ties on the centrality side), at the cost of fewer observed nodes per bin. The default $K_{\max}$ is $\min(20, \lfloor N_{\text{obs}}/5 \rfloor)$, the default $K_{\min}$ is 4. Users may override with an explicit integer when manual control is desired.
- J: number of E/I bins (default 3, with semantic categories internal-leaning [E/I < -0.33], balanced [$-0.33 \leq$ E/I ≤ 0.33], broker-leaning [E/I > 0.33])

3.2 Feasibility of ρ

Not every $\rho \in (-1, 1)$ is structurally achievable on a given network at a given missingness rate. Two distinct ceilings constrain the realized correlation:

Rate-bounded absolute ceiling. Under Kendall's τ_b between a binary indicator (missingness, with proportion p) and a continuous variable (centrality), the realized correlation is structurally bounded by

$$|\tau_b|_{\max} = \sqrt{\frac{2 \cdot k \cdot (n-k)}{n \cdot (n-1)}} \approx \sqrt{2 \cdot p \cdot (1-p)}$$

where k is the number of missing nodes and $p = k/n$. This bound depends only on the missingness rate, not on the network structure. At $p = 0.10$, $|\tau_b|_{\max} \approx 0.42$; at $p = 0.25$, $|\tau_b|_{\max} \approx 0.61$; at $p = 0.50$, $|\tau_b|_{\max} \approx 0.71$. The framework cannot produce realized ρ exceeding this absolute ceiling no matter how the sampling is configured.

Practical tie-bounded ceiling. On networks where centrality has many ties (most unweighted networks, where multiple nodes share the same degree), the achievable $|\tau_b|$ is further reduced below the absolute ceiling. Centrality ties contribute to the denominator of Kendall τ_b, lowering the maximum that any sampling configuration can reach. The practical ceiling is network-specific and depends on the centrality distribution's tie structure.

Feasibility check. The framework provides a public function `feasible_rho_range` that probes the sampling mechanism at extreme configurations to estimate the practical $[\rho_{\min}, \rho_{\max}]$ achievable on a given network at a given rate. The wrapper `reconstruct_network` calls this function at startup and emits a warning if the user's ρ falls outside the practical range, with three tiers of warning intensity (silent, note, prompt) depending on the degree of infeasibility.

What happens when ρ exceeds the practical ceiling. The framework does not silently clip or rescale. Instead, the Phase 1 sampler's bisection on the mixing scalar b saturates at the extreme value, and the resulting *realized* ρ is whatever the network can structurally produce. Phase 2's bin-distribution solver then honors the realized ρ rather than the user's nominal request. Diagnostic output reports both nominal and realized values, allowing the user to interpret credible intervals appropriately.

Asymmetry between $+\rho$ and $-\rho$. Under Kendall τ_b with rank-based sampling weights (see Step 5), feasibility ceilings are typically symmetric: $|\rho_{\min}| \approx |\rho_{\max}|$. This is a substantive improvement over Pearson correlation, where heavy-tailed networks produced dramatically asymmetric ceilings (e.g., $\rho_{\max} \approx +0.4$ but $\rho_{\min} \approx -0.03$ on Marvel under Pearson). The `is_asymmetric` flag in `feasible_rho_range` output retains a diagnostic for unusual cases but rarely fires under Kendall.

3.3 Weight Semantics and Transformation

The framework assumes edge weights are non-negative integer counts representing interactions or analogous additive quantities where higher weight indicates a stronger tie. Networks where weights represent distances, dissimilarities, or other "lower = stronger" quantities must be transformed to a count-like scale before processing.

When `weight_semantics = "distance"` is set, the framework applies one of the following transformations in a preprocessing step:

- **Scaled reciprocal (default):** $w' = \mathrm{round}(c/w)$ with $c = \max(w)$. Requires distances to be strictly positive; zero-distance edges, if any, take the maximum observed transformed weight.
- **Max-minus-distance:** $w' = \mathrm{round}(\max(w) - w)$. Linear inversion; suitable when distances have a meaningful upper bound. Distance-zero maps to maximum count; maximum distance maps to zero (treated as no edge).
- **Exponential decay:** $w' = \mathrm{round}(c \cdot \exp(-w/\tau))$ for user-supplied decay parameter τ; c defaults to a value chosen so that the median transformed weight is approximately 10.

After transformation, all subsequent steps operate on the transformed network. Credible intervals returned by the framework apply to the transformed network, not the original distances. Users wishing to make inferences about original-scale metrics must back-transform results carefully, recognizing that nonlinear transformations may distort the interval structure. For rank-based measures (e.g., who has highest degree centrality), rank ordering is often preserved across reasonable transformations; for distance-dependent measures (e.g., closeness, betweenness on weighted shortest paths), the transformation fundamentally changes what the measure represents.

3.4 The Bin-Exchangeability Assumption

The framework treats all observed nodes within a bin as exchangeable. When sampling edges for added nodes, bin-level connection probabilities and weights are used rather than node-level. This is the framework's central simplifying assumption.

By default, binning is two-dimensional: each node is characterized by both its degree rank and its E/I index. This recognizes that two nodes with the same degree may play fundamentally different structural roles. Within each (degree-bin $\times$ E/I-bin) cell, individual differences in connection patterns are still averaged out; to the extent that within-cell structural variation matters for the user's measure of interest, credible intervals may be miscalibrated. The framework is best suited to networks where degree and E/I together capture the dominant structural patterns; networks dominated by attribute homophily, geographic structure, or community structure orthogonal to degree-and-brokerage remain difficult cases.

3.5 Structural-Role Binning via the E/I Index

The framework always uses the E/I index as a second binning dimension alongside degree. The E/I index measures how much a node's ties cross community boundaries, distinguishing brokers (high E/I) from clustered hubs (low E/I) even when they have the same degree.

Definition. For each node i,

$$\mathrm{E/I}_i = \frac{E_i - I_i}{E_i + I_i},$$

where E_i is the count (or weight, for weighted networks) of node i's external ties (crossing community boundaries) and I_i is the count (or weight) of its internal ties. The index ranges from -1 (purely internal) to $+1$ (purely external).

Community detection. If user-supplied community labels are not provided, the framework runs Louvain modularity optimization on G_{obs}. For directed networks, the network is symmetrized for community detection only; the directed structure is retained for all other steps. For weighted networks, Louvain uses edge weights directly. Community detection runs once at setup and is held fixed across all replicates.

Fallback behavior. If community detection returns a single community, or if E/I has insufficient variance to support J bins with at least three nodes per bin, the framework issues a diagnostic message and falls back to degree-only binning for the remainder of the run. This handles edge cases gracefully: networks with no meaningful community structure, very small networks where community detection is unreliable, and networks where all nodes have similar bridging behavior. The fallback is automatic; no user intervention is required. The binning mode used (2D or fallback to 1D) is reported in diagnostics.

Two-dimensional binning. When E/I binning is active, each observed node is assigned both a degree bin (1 to K) and an E/I bin (1 to J). The rank-rank connectivity matrix is indexed by (degree-bin, EI-bin) cell pairs. Because per-cell estimates can be noisy when cells contain few nodes, the framework applies Laplace smoothing to the connection probabilities P in all cases.

Bin assignment for added nodes. Added nodes are assigned degree-bins via the ρ-weighted distribution. Their E/I bins are drawn from the empirical conditional distribution of E/I bin given the assigned degree bin in G_{obs}. Central brokers are therefore sampled when central added nodes are drawn, in proportion to how common brokers are at high centrality in the observed network. Users with specific prior beliefs about brokerage-missingness correlation (distinct from centrality-missingness) may extend the framework with a separate $\rho_{\text{E/I}}$ parameter; this is left as an advanced extension.

3.6 Nominated Non-Respondents and the Reciprocity Matrix

In many observation processes, the analyst has explicit knowledge of some unobserved nodes because they appear as targets of edges from observed nodes. In survey settings, these are non-respondents who were nominated by respondents; in surveillance settings, they are actors named in observed activity reports; in social media samples, they are accounts mentioned by sampled users. The framework refers to these as *nominated non-respondents* (or, more generally, partially observed nodes).

The user identifies these nodes through the optional input `partially_observed_nodes`. When supplied, the framework treats them differently from the synthetic "added" nodes generated by Stage 1: they are placed in the network using their observed incoming edges, and their outgoing edges are imputed using a rank-conditional reciprocity model.

The reciprocity matrix R (directed networks only). For each ordered pair of cells (c, c'), define

$$R_{c,c'} = \Pr\left(j \to i \text{ exists} \,\middle|\, i \to j \text{ exists},\ i \in c,\ j \in c'\right).$$

In words: $R_{c,c'}$ is the probability that a forward edge from a node in cell c to a node in cell c' is reciprocated by a reverse edge from c' back to c. The matrix is estimated empirically from respondent-respondent dyads in G_{obs} with Laplace smoothing:

$$R_{c,c'} = \frac{|\{(i,j) : i \in c, j \in c', w(i,j) > 0, w(j,i) > 0\}| + 1}{|\{(i,j) : i \in c, j \in c', w(i,j) > 0\}| + 2}.$$

Unlike the unconditional matrix P, the reciprocity matrix R is fundamentally asymmetric: $R_{c,c'}$ and $R_{c',c}$ can be very different in networks with structurally asymmetric tie patterns (e.g., hub-and-spoke or core-periphery networks).

Generalization of probabilistic imputation. The matrix R generalizes the single global reciprocity rate used in the probabilistic imputation of Smith et al. (2022). Where the prior approach used one value for all dyads (e.g., 0.25 in their illustrative example), the framework estimates a separate value for each rank-rank cell pair. With broker detection enabled, R also captures characteristic asymmetries between brokers and clustered hubs that a single rate would miss.

Bin assignment for nominated non-respondents. Each nominated non-respondent is placed in degree and E/I bins based on its *observed* incoming edges. This means the bin assignment may under-estimate the node's true centrality (since outgoing edges are missing), but it uses only information actually available in G_{obs}. Adjusting for under-observation is left as an advanced extension.

Undirected networks. The reciprocity matrix R is undefined for undirected networks. For undirected G_{obs}, the framework treats edges between respondents and nominated non-respondents as fully observed (taking the respondent's report at face value), and uses the unconditional matrix P to impute edges between pairs of nominated non-respondents. This is a known limitation: undirected networks dominated by edges that respondents failed to report (measurement error) are less well-handled by the framework.

3.7 Edge Representation Convention

The framework operates on networks that may be directed or undirected, weighted or unweighted:

- Directed networks: centrality combines in-degree and out-degree; rank-rank matrices index ordered pairs; added-node edges are sampled separately for outgoing and incoming directions.
- Undirected networks: centrality is total degree; rank-rank matrices are symmetric and index unordered pairs; added-node edges are sampled once per pair.
- Weighted networks: edge-weight reversal (Stage 2) is available; added-node edge weights are drawn from Poisson distributions.
- Unweighted networks: Stage 2 is skipped; added-node edges have weight 1 by construction; the rank-rank weight matrix is unnecessary.

Self-loops are ignored throughout. Multigraphs must be collapsed to weighted single-edge form before input.

3.8 Step 0 — Preprocessing

Step 0a: Weight transformation (if applicable). If `weight_semantics = "distance"`, apply the selected transformation to G_{obs}. All subsequent steps operate on the transformed network. Transformation parameters are recorded for output metadata.

Step 0b: Resolve missingness parameters. If G_{obs} (post-transformation, if applicable) is unweighted, set $\pi_{\text{edge}} = 0$ regardless of user input, issue a warning if the user supplied a non-zero π_{edge}, and require $\pi_{\text{node}} > 0$. Otherwise: if π_{node} and/or π_{edge} are explicitly provided, use them as given (defaulting the unspecified one to 0); else if π_{total} is provided, set $\pi_{\text{node}} = \pi_{\text{total}}$ and $\pi_{\text{edge}} = 0$. Validate that $\pi_{\text{node}}, \pi_{\text{edge}} \in [0, 1]$ and at least one is positive.

3.9 Setup Phase

Step 1: Compute observed centrality. For directed G_{obs}, set degree centrality as in-degree plus out-degree per node. For undirected G_{obs}, set centrality as degree. Rank nodes by centrality; ties broken consistently.

Step 1.5: Detect community structure and compute E/I. If user-supplied community labels are provided, use them. Otherwise, apply Louvain to G_{obs} (symmetrizing first if directed; using weights if weighted). Check whether the community structure is meaningful: more than one community detected, and E/I values supporting J bins with at least three nodes each. If meaningful, compute E/I$_i$ for each observed node and rank by E/I. If not, fall back to degree-only binning for the remainder of the run and log a diagnostic.

Step 2: Bin observed nodes. Partition observed nodes into K equally-sized degree bins, with bin 1 the most peripheral and bin K the most central. If 2D binning is active, additionally partition into J E/I bins using the semantic thresholds (-0.33, $+0.33$) for $J = 3$, or equally-sized rank bins for other J.

Step 3: Compute the rank-rank connectivity matrix P. For each cell pair (c, c') (ordered for directed networks, unordered for undirected), compute

$$p_{c,c'} = \frac{\text{count of edges from cell } c \text{ to cell } c' + 1}{\text{total ordered pairs between cells} + 2}$$

(Laplace-smoothed) and, for weighted networks,

$$w_{c,c'} = \text{mean weight of edges from } c \text{ to } c'\text{, conditional on edge presence.}$$

For unweighted networks, only P is computed. If a cell pair has no observed edges, $w_{c,c'} = 0$ and any sampled edge from that pair is assigned weight 1. The matrix is computed over all respondent-respondent dyads in G_{obs} (excluding any partially observed nodes, which are placed using their own observed edges separately).

Step 3.5: Compute the reciprocity matrix R (directed networks only). For each ordered cell pair (c, c'), compute the Laplace-smoothed conditional reciprocity rate

$$R_{c,c'} = \frac{|\{(i,j) : i \in c, j \in c', \, w(i,j) > 0, \, w(j,i) > 0\}| + 1}{|\{(i,j) : i \in c, j \in c', \, w(i,j) > 0\}| + 2}.$$

Computed only over respondent-respondent dyads. The matrix R is generally asymmetric: $R_{c,c'} \neq R_{c',c}$ in most networks. This step is skipped for undirected networks, where R is undefined.

Step 4: Determine the number of nodes to add (non-nominated case). Let N_{nom} be the number of partially observed (nominated non-respondent) nodes supplied by the user. The framework adds N_{add} *additional* synthetic nodes representing non-nominated non-respondents:

$$N_{\text{add}} = \text{round}\left(\frac{\pi_{\text{node}} \cdot (N + N_{\text{nom}})}{1 - \pi_{\text{node}}}\right) - N_{\text{nom}}.$$

If $N_{\text{nom}} = 0$, this reduces to $N_{\text{add}} = \text{round}(\pi_{\text{node}} \cdot N/(1 - \pi_{\text{node}}))$ as in the original formulation. If $\pi_{\text{node}} = 0$, skip both Stage 0.5 and Stage 1; the framework reduces to Stage 2 only. The total number of nodes in each replicate's true-network estimate is $N + N_{\text{nom}} + N_{\text{add}}$.

Step 5: Determine the bin-assignment distribution for added nodes. Find a distribution q over degree bins $\{1,\ldots,K\}$ such that drawing N_{add} bins from q yields realized Kendall τ_b correlation ρ between the bin index and the missingness indicator across all $N+N_{\text{add}}$ nodes. Use the logistic form

$$q_b \propto \exp(\beta \cdot \tilde{b}),$$

where $\tilde{b}$ is the standardized bin index. Solve numerically for β such that the induced Kendall τ_b matches ρ:

- $\rho > 0$: $\beta > 0$, q skews toward high bins
- $\rho < 0$: $\beta < 0$, q skews toward low bins
- $\rho = 0$: $\beta = 0$, q uniform across bins

Analytic Kendall τ_b formula. Because bins are ordinal and missingness is binary, Kendall τ_b has a closed-form expression as a function of β, K, N, and N_{add}. Only cross-pairs (one observed node, one added node) contribute to concordant/discordant counts; same-status pairs are tied on the indicator. With observed nodes uniformly distributed across bins (N/K per bin) and added nodes distributed by q:

$$C = \frac{N}{K}\cdot N_{\text{add}}\cdot\sum_{k=1}^{K} q_k\cdot(k-1), \qquad D = \frac{N}{K}\cdot N_{\text{add}}\cdot\sum_{k=1}^{K} q_k\cdot(K-k),$$

$$T_x = \frac{N\cdot N_{\text{add}}}{K}, \qquad T_y = \frac{N^2(K-1)}{2K} + \frac{N_{\text{add}}^2}{2}\left(1-\sum_k q_k^2\right),$$

$$\tau_b = \frac{C-D}{\sqrt{(C+D+T_x)(C+D+T_y)}}.$$

This is the function being numerically inverted to solve for β. The formula is monotone increasing in β, so a straightforward bisection on $\beta \in [-50, 50]$ converges in a small number of iterations to tight tolerance.

***K*-dependent ceiling.** The maximum achievable $|\tau_b|$ from Step 5's mechanism depends on K: larger K reduces the bin-side ties T_x and lifts the ceiling closer to the rate-bounded absolute $\sqrt{2p(1-p)}$. Smaller K increases bin ties and lowers the ceiling. This is one reason the framework's default K is chosen by `find_optimal_K` (Section 3.11): for a user-specified ρ near the rate-bounded ceiling, only sufficiently large K can avoid `:ceiling_hit` in the bisection.

β-solver status. The solver returns one of three statuses: `:converged` (bisection found β within tolerance of the target), `:ceiling_hit` (target exceeds the K-dependent achievable maximum; β saturates at the bracket endpoint), or `:failed_other` (degenerate case, e.g., $N_{\text{add}} = 0$ with nonzero target). The status is recorded in diagnostics.

In 2D binning mode, also compute the empirical conditional distribution of E/I bin given degree bin in G_{obs}, for use in Step 6.

3.10 Replicate Generation Phase

For each replicate $b \in \{1,\ldots,B\}$:

Step 6: Draw bin assignments for added nodes (if $\pi_{\text{node}} > 0$). For each of N_{add} added nodes, draw a degree-bin assignment from q. In 2D binning mode, draw an E/I-bin assignment from the conditional distribution given the assigned degree bin.

Step 6.5: Stage 0.5 — impute edges for nominated non-respondents (if $N_{\text{nom}} > 0$). This stage processes nominated non-respondents using their observed incoming edges and the reciprocity matrix R. Each nominated non-respondent E enters this stage with its known incoming edges from respondents already in place; only the outgoing and additional edges need imputation.

Directed case. For each nominated non-respondent E:

- **Reverse edges to respondents who nominated E.** For each respondent A with an observed forward edge $A \to E$: draw $\text{Bernoulli}(R_{c_A,c_E})$ for the reverse edge $E \to A$, where c_A and c_E are the cell assignments of A and E. If yes, assign weight 1 (unweighted) or draw weight from $\text{Poisson}(w_{c_E,c_A})$ with minimum 1 (weighted).

- **Outgoing edges to respondents who did not nominate E.** For each respondent A with no observed edge to or from E: draw $\text{Bernoulli}(p_{c_E,c_A})$ for the edge $E \to A$. If yes, assign weight as above. This uses the unconditional matrix P because there is no forward edge to condition on.

- **Edges between pairs of nominated non-respondents.** For each pair (E, F) of nominated non-respondents: draw both $E \to F$ and $F \to E$ independently via $\text{Bernoulli}(p)$ using the appropriate cells of P.

Undirected case. The reciprocity matrix R does not apply. For each nominated non-respondent E:

- **Observed edges from respondents to E are taken as given** (no imputation needed).

- **Edges between pairs of nominated non-respondents.** For each unordered pair $\{E, F\}$: draw $\text{Bernoulli}(p)$ using the appropriate cell of P; if yes, assign weight as above.

Step 7: Stage 1 — add edges incident to non-nominated added nodes. *Directed case.* For each added node v (synthetic, non-nominated) with bin assignment(s):

- **Outgoing edges from v to observed and nominated nodes.** For each such target node j: draw $\text{Bernoulli}(p)$ using the appropriate cell pair. If yes, assign weight 1 (unweighted) or draw weight from $\text{Poisson}(w)$ with minimum 1 (weighted).

- **Incoming edges from observed and nominated nodes to v.** Same logic with reversed direction.

- **Edges between added nodes.** Same logic, both endpoints added.

Undirected case. For each added node v:

- **Edges between v and observed or nominated nodes.** For each such node j: draw $\text{Bernoulli}(p)$ using the appropriate cell pair. If yes, assign weight as above.

- **Edges between pairs of added nodes.** For each unordered added-node pair: draw $\text{Bernoulli}(p)$; if yes, assign weight as above.

The result is G_{aug}: G_{obs} (post-transformation if applicable) plus nominated non-respondents with their imputed edges (from Stage 0.5) plus N_{add} synthetic nodes with their sampled edges. No new edges are created between two original respondent nodes. No self-loops are sampled.

Step 8: Stage 2 — reverse the edge-weight removal process (if $\pi_{\text{edge}} > 0$). Skipped entirely if G_{obs} is unweighted or $\pi_{\text{edge}} = 0$. Let W_{aug} be the total edge weight in G_{aug}. Compute

$$W_{\text{add}} = \text{round}\left(\frac{\pi_{\text{edge}} \cdot W_{\text{aug}}}{1 - \pi_{\text{edge}}}\right).$$

Sample W_{add} weight units, each assigned to an existing edge with probability proportional to current weight, with replacement; increment the corresponding edge weights. This inverts Chapter 3's removal mechanism: where the noise model removed weight units proportional to current weight, we add them back the same way. The result is G_b.

Step 9: Compute measure(s) on the replicate. For network-level measures, compute $m(G_b)$. For node-level measures, compute $m_i(G_b)$ for each respondent node i in G_{obs} and (optionally) for each nominated non-respondent. Synthetic added nodes are nuisance variables introduced for structural realism; their measure values are not reported. Nominated non-respondents have partial observed data plus imputed edges, so node-level credible intervals for these nodes reflect both observational and imputation uncertainty.

3.11 The Three-Prior Algorithmic Contract

The framework operates as an algorithmic contract that honors three priors on each replicate:

1. **Prior 1 (proportion missing).** The user supplies π_{node}. The framework honors this via the arithmetic identity $N_{\text{add}}/(N + N_{\text{add}}) = \pi_{\text{node}}$ (modulo integer rounding). Verified by the inputs to `_determine_n_add`.

2. **Prior 2 (centrality-missingness correlation).** The user supplies ρ. The framework honors this via the β solver in Step 5, which finds β such that the analytic Kendall τ_b between bin index and missingness equals ρ within tolerance. When ρ exceeds the K-dependent ceiling, the solver returns `:ceiling_hit`, and the realized ρ is the maximum achievable at the given K, N, and N_{add}. Both `:converged` and `:ceiling_hit` are contract-compliant responses; the framework's job is to tell the user honestly what it could and could not achieve.

3. **Prior 3 (E/I distribution conditional on degree).** This prior is *inferred* from G_{obs} rather than user-supplied. The framework computes the empirical conditional distribution $P(\text{E/I bin} \mid \text{degree bin})$ from observed nodes and uses it to assign E/I bins to added nodes (with uniform fallback for empty degree bins). The contract is that added nodes' E/I distribution matches G_{obs}'s empirical conditional row-by-row.

What the contract guarantees and what it does not. The contract guarantees the framework will honor each prior *algorithmically*: the realized values across replicates will match the priors within tolerance, or the framework will report exactly where it could not (via `:ceiling_hit` status flags). The contract does *not* guarantee that the user's priors are *correct*: a user who specifies $\rho = 0$ when the truth is $\rho = 0.5$ will receive miscalibrated credible intervals, but the framework will honor $\rho = 0$ as requested.

Automatic K selection. The function `find_optimal_K` searches for the largest K that satisfies all three priors. Because the Phase 2 analytic Kendall ceiling depends on K (larger K permits larger realized ρ), users specifying ρ near the rate-bounded ceiling benefit from larger K to avoid `:ceiling_hit`. The search is preceded by a feasibility check via `feasible_rho_range`: if ρ is structurally infeasible, the function returns immediately with `:rho_infeasible` rather than searching K values that cannot succeed.

3.12 Aggregation Phase

Step 10: Summarize the posterior sample. For each network-level measure m, the posterior sample is $\{m(G_b)\}_{b=1}^{B}$. Default summaries: posterior median and 95% credible interval [quantile(0.025), quantile(0.975)]. For each node-level measure and each original observed node i, the posterior sample is $\{m_i(G_b)\}_{b=1}^{B}$, with per-node default summaries identical in form.

3.13 Output

The framework returns:

- **Primary output:** the full posterior sample (B values per network-level measure, or B values per observed node per node-level measure)
- **Default summaries:** posterior median and 95% credible interval; alternative credible levels and other summaries can be computed from the sample
- **Reference values:** $m(G_{\text{obs}})$ on the observed (post-transformation if applicable) network
- **Diagnostics:**
  - Nominal vs. realized ρ (Kendall τ_b) per replicate, with summary across replicates
  - Nominal vs. realized missingness rate per replicate
  - β solver status distribution (`:converged`, `:ceiling_hit`, `:failed_other`) across replicates
  - Feasibility output from `feasible_rho_range`: practical $[\rho_{\min}, \rho_{\max}]$ at the user's missingness rate, MC standard errors, asymmetry flag
  - K selected by `find_optimal_K` (if `K = :auto`) and per-K test trace
  - Transformation parameters used (if weight semantics required transformation)
  - Community structure used and E/I distributions
  - Binning mode (2D or fallback to 1D) and any fallback reason
  - Per-replicate degeneracy flags (giant-component fraction, empty bins, etc.) for downstream filtering

4 Key Assumptions

The framework makes the following assumptions, which users should consider when interpreting results:

1. **Edge presence among respondent nodes is fully observed.** Any two respondent nodes (i.e., observed nodes that are not partially observed nominated non-respondents) either interacted (and we recorded it) or did not. For weighted networks, Stage 2 may add weight to existing edges but does not create new edges among respondents. For unweighted networks, no edge modifications among respondents occur at all. Edges involving nominated non-respondents are partially observed and handled by Stage 0.5; edges involving non-nominated added nodes are handled by Stage 1.

2. **Rank-rank connection structure in G_{obs} is representative of the true network.** The probability that a node in cell c connects to a node in cell c' in the truth equals the empirical $p_{c,c'}$ from G_{obs} in expectation.

3. **Rank-conditional reciprocity behavior is representative.** For directed networks, the empirical reciprocity rate $R_{c,c'}$ observed among respondent-respondent dyads is representative of reciprocity behavior involving nominated non-respondents at the same rank-rank cell pair. This generalizes the assumption in Smith et al. (2022) that a single global reciprocity rate could be used for imputation; the framework's assumption is that reciprocity varies by rank-rank cell but is otherwise constant.

4. **ρ is a genuine prior, not estimated from $G_{\mathbf{obs}}$.** Users supply ρ based on domain knowledge about the observation process, not by looking at the data. The framework interprets ρ internally as Kendall's τ_b between the missingness indicator and node centrality. When the user's ρ exceeds the practical feasibility ceiling (Section 3.2), the framework operates at the ceiling and reports both nominal and realized values via diagnostics; the contract is algorithmic compliance with what is structurally achievable, not silent overrides of the user's request.

5. **Bin membership, not node identity, governs sampling.** All observed nodes in the same cell are treated as exchangeable. This is the framework's central simplifying assumption (see Section on bin exchangeability).

6. **Observed-node properties are fixed across replicates.** Degree ranks, E/I ranks, and community labels are computed once on G_{obs} and held constant. Added nodes are assigned bins but do not enter the rankings that define bins. Nominated non-respondents are binned based on their observed incoming edges only.

7. **Edge weights (when present) are integer-valued counts of interactions or analogous higher-is-stronger additive quantities.** Distance, dissimilarity, or lower-is-stronger semantics are not supported directly; users must specify a transformation method, after which credible intervals apply to the transformed scale.

8. **Default behavior treats missingness as node-level.** Edge-level error on observed nodes (π_{edge}) is opt-in via the advanced interface or non-zero explicit specification, and is unavailable for unweighted networks.

9. **Community structure derived from $G_{\mathbf{obs}}$ reflects the true community structure.** Louvain output (or user-supplied labels) is treated as fixed and correct. To the extent the true community structure differs from what is detected in G_{obs} — particularly because missingness may obscure community boundaries — the E/I index becomes a noisier signal of bridging behavior.

10. **Higher-order structural patterns beyond degree and E/I are not directly modeled.** Triadic closure, attribute homophily, and other patterns appear in replicate networks only to the extent that they correlate with the binning variables. Networks dominated by patterns orthogonal to the binning are less faithfully reproduced.

11. **The output is an approximate Bayesian posterior.** The framework produces samples from a distribution consistent with the user's prior, but does not derive from an explicit likelihood and prior density formulation.

5 What the Framework Does Not Model

The following are explicitly outside the scope of this framework:

- Creation of edges between two respondent nodes (the framework assumes edge *presence* among respondents is correctly observed; only *weights* are uncertain, and only for weighted networks). Edges involving nominated non-respondents *are* created/imputed by Stage 0.5.

- Respondent measurement error: cases where a respondent failed to report an edge to another respondent that actually exists. The framework treats all respondents as accurate reporters of their own edges.
- Non-uniform edge-weight missingness (Stage 2 is MCAR at the weight-unit level among existing edges)
- Spurious edges in the observed network (the framework assumes observed structure is correct, not over-reported)
- Higher-order structural patterns beyond degree and E/I (triadic closure, attribute homophily, community structure orthogonal to the bins are captured only incidentally)
- Changes in node ranks or community memberships due to the augmentation process (all are fixed at G_{obs})
- Self-loops in added-node sampling (excluded by convention)
- Multigraphs in input (must be collapsed to weighted single-edge form before running)
- Edge-weight missingness for unweighted networks (structurally impossible)
- Native handling of distance, dissimilarity, or lower-is-stronger weight semantics (must be transformed to count-like scale via the framework's preprocessing step)
- Separate prior on brokerage-missingness correlation (ρ governs degree-missingness; E/I assignment is conditional on degree)
- Uncertainty in the detected community structure (Louvain output is treated as fixed)
- Reciprocity-based imputation for undirected networks (the reciprocity matrix R is undefined; nominated non-respondents in undirected networks are handled with the unconditional rank-rank matrix P only)
- Silent enforcement of structurally infeasible ρ priors. When the user's ρ exceeds the rate-bounded or tie-bounded ceiling, the framework operates at the achievable ceiling, reports the realized ρ alongside the nominal, and emits a diagnostic. It does not silently rescale or clip the user's input.

6 Next Steps

This document specifies the algorithm in sufficient detail for Python implementation. Subsequent work will include:

1. Implementation of the framework in Python, with attention to computational efficiency for the B replicates and the rank-rank matrix construction
2. Empirical validation on simulated networks with known ground truth, varying network type, missingness mechanism, and ρ
3. Validation on real network data, including the covert and survey-based networks used in prior work
4. Comparison against the Chapter 4 bootstrap methods to assess whether the framework's intervals are better calibrated under non-random missingness
5. Extension to multidimensional (multiplex and/or dynamic) networks, building on the tensor representations developed in Chapter 3

References

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